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(54) **STRAWBERRY PLANT VARIETY NAMED 'DRISSTRAWSEVENTYEIGHT'**

(50) Latin Name: ***Fragaria* x *ananassa***
Varietal Denomination: **DrisStrawSeventyEight**

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USPC **Plt./208**

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(57) **ABSTRACT**

A new and distinct variety of strawberry plant named 'DrisStrawSeventyEight', selected for its early high yield without long gaps, fruit size, fruit eating quality, well exposed fruit, and compact plant architecture, is disclosed.

5 Drawing Sheets

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STRAWBERRY PLANT VARIETY NAMED 'DRISSTRAWSEVENTYEIGHT'

Botanical classification: *Fragaria* x *ananassa*.

Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawSeventyEight'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawSeventyEight'.

Strawberry plant variety 'DrisStrawSeventyEight' originated from a cross between the female parent 'DrisStrawForty' (U.S. Plant Pat. No. 25,747) and a proprietary male parent that was selected from the cross of 'Pelican' (unpatented) x '29Q268' (unpatented) (hereinafter 'Pelican x 29Q268', unpatented). Progeny plants from this cross of 'DrisStrawForty' x 'Pelican x 29Q268', including 'DrisStrawSeventyEight', were asexually propagated via stolons in Ciudad Guzman, Jalisco, Mexico in March of 2012. Strawberry plant variety 'DrisStrawSeventyEight' was later specifically identified and selected in Tangancicuaro, Michoacan, Mexico in October of 2012.

'DrisStrawSeventyEight' was subsequently asexually propagated via stolons, and underwent further testing at a farm in Tangancicuaro, Michoacan, Mexico for six years (2012 to 2018). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons.

'DrisStrawSeventyEight' exhibits the following distinguishing characteristics when grown under normal horticultural practices in Tangancicuaro, Michoacan, Mexico:

1. Inflorescence above foliage;
2. Obtuse shape of base of terminal leaflet; and
3. Convex shape in cross section of terminal leaflet.

'DrisStrawSeventyEight' was selected for its early high yield without long gaps, fruit size, fruit eating quality, well exposed fruit, and compact plant architecture.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs which show fruit of the plant, flowers, leaves, and the plants. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. The photographs are of plants that are seven months old.

FIG. **1** illustrates whole fruit of variety 'DrisStrawSeventyEight'.

FIG. **2** illustrates longitudinal sections of fruit of variety 'DrisStrawSeventyEight'.

FIG. **3** shows the upper and lower surfaces of flowers of variety 'DrisStrawSeventyEight'.

FIG. **4**A shows the upper surface of a leaf of variety 'DrisStrawSeventyEight'. FIG. **4**B shows the lower surface of a leaf of variety 'DrisStrawSeventyEight'.

FIG. **5** shows plants of variety 'DrisStrawSeventyEight'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawSeventyEight'. The data which define these characteristics is based on observations taken in Tangancicuaro, Michoacan, Mexico from 2012 to 2018. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawSeventyEight' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawSeventyEight' was taken from plants that were seven months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The RHS Colour Chart of The Royal Horticultural Society of London (RHS) (2007 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:

- *Species.*—*Fragaria* x *ananassa.*
- *Common name.*—Strawberry.
- *Denomination.*—'DrisStrawSeventyEight'.

Parentage:

- *Female parent.*—The variety 'DrisStrawForty' (U.S. Plant Pat No. 25,747).
- *Male parent.*—The proprietary variety 'Pelican x 29Q268' (unpatented).

Plant:

- *Height.*—23.0 cm.
- *Diameter.*—36.4 cm.
- *Number of crowns per plant.*—3 .
- *Growth habit.*—Semi-upright.

Stolon:

- *Average number of daughter plants per square foot.*—17.
- *Diameter at bract.*—3.74 mm.
- *Anthocyanin coloration.*—Weak.
- *Anthocyanin color.*—RHS 42A (Vivid reddish orange).

Leaf:
- *Number of leaflets.*—Three only.
- *Color of upper surface.*—RHS 136B (Dark yellowish green).
- *Variegation.*—Absent.
- *Terminal Leaflets.*—Length: 9.5 cm. Width: 4.4 cm. Length/width ratio: 2.2. Number of teeth/terminal leaflet: 22. Shape of base: Obtuse. Margin: Serrate to crenate. Shape in cross section: Convex.
- *Petiole.*—Length: 13.4 cm. Diameter: 3.23 mm. Attitude of hairs: Slightly outwards. Bract frequency (number present on each petiole): 2.
- *Petiolule.*—Length: 14.10 mm. Diameter: 2.43 mm.
- *Stipule.*—Length: 3.2 cm. Width: 17.60 mm. Anthocyanin coloration: Weak. Anthocyanin color: RHS 61D (Deep purplish pink).

Inflorescence:
- *Position in relation to foliage.*—Above.
- *Pedicel.*—Attitude of hairs: Slightly outwards.
- *Flower.*—Flower diameter (petal tip to petal tip on non-flattened flower): 27.60 mm. Arrangement of petals: Overlapping. Stamen: Present. Typical and observed number of flowers per plant: 5.70.
- *Petal.*—Length: 13.30 mm. Width: 14.00 mm. Length/width ratio: 1.0. Typical and observed petal number: 6. Color of upper side: RHS 11D (Pale yellow).
- *Calyx.*—Diameter (sepal tip to sepal tip, measured on back of flower): 43.50 mm.
- *Sepal.*—Length (sepal tip to point of attachment to receptacle): 21.80 mm. Width: 11.70 mm. Typical and observed sepal number: 11.

Fruit:
- *Length.*—42.14 mm.
- *Width.*—40.93 mm.
- *Length/width ratio.*—1.0.
- *Weight (individual fruit).*—26.9 g.
- *Fruit hollow length.*—20.08 mm.
- *Fruit hollow width.*—13.79 mm.
- *Fruit hollow length/width ratio.*—1.5.
- *Shape.*—Cordate.
- *Color.*—RHS 45A (Vivid red).
- *Position of achenes.*—Level with surface.
- *Position of calyx attachment.*—Inserted.
- *Attitude of sepals.*—Upwards.
- *Color of flesh (excluding core).*—RHS 40B (Vivid reddish orange).
- *Color of core.*—RHS 37A (Strong yellowish pink).
- *Fruiting truss.*—Length (from crown to base of terminal flower or fruit): 21.0 cm. Diameter (at base of truss): 5.93 mm. Number of berries per fruiting truss: 7.1

Production:
- *Flowering interval.*—Mid-September to mid-March.
- *Harvest interval.*—Early October to mid-March.
- *Type of bearing.*—Partially remontant.
- *Productivity.*—32,456 kg to 51,329 kg of fruit per hectare per season from seven-month-old plants when grown in Tangancicuaro, Michoacan, Mexico.

Resistance to abiotic stress, pests, and diseases:
- *Heat.*—Moderately resistant.
- *Two-spotted spider mite (Tetranychus urticae).*—Moderately susceptible.

- *Botrytis fruit rot (Botrytis cinerea).*—Moderately resistant.
- *Powdery mildew (Podosphaera macularis).*—Moderately resistant.
- *Xanthomonas fragariae.*—Moderately susceptible.

COMPARISON WITH PARENTAL AND COMMERCIAL VARIETIES

'DrisStrawSeventyEight' differs from the female parent 'DrisStrawForty' (U.S. Plant Pat. No. 25,747) in that plants of 'DrisStrawSeventyEight' produce fruit with better flavor and better shelf life than plants of 'DrisStrawForty'. Additionally, plants of 'DrisStrawSeventyEight' have a semi-upright growth habit and a partially remontant type of bearing, while plants of 'DrisStrawForty' have a spreading growth habit and a not remontant type of bearing. Moreover, terminal leaflets of 'DrisStrawSeventyEight' have an obtuse shape of base and a convex shape in cross section, while terminal leaflets of 'DrisStrawForty' have a rounded shape of base and a concave shape in cross section.

'DrisStrawSeventyEight' differs from the proprietary male parent 'Pelican x 29Q268' (unpatented) in that fruit of 'DrisStrawSeventyEight' is firmer and more red than fruit of 'Pelican x 29Q268'.

'DrisStrawSeventyEight' differs from the commercial variety 'Driscoll Osceola' (U.S. Plant Pat. No. 15,752) in that 'DrisStrawSeventyEight' has inflorescence above foliage, while 'Driscoll Osceola' has inflorescence level with to above foliage. Additionally, 'DrisStrawSeventyEight' has an obtuse shape of base of terminal leaflet and a convex shape in cross section of terminal leaflet, while 'Driscoll Osceola' has a rounded shape of base of terminal leaflet and a concave to slightly concave shape in cross section of terminal leaflet. Moreover, 'DrisStrawSeventyEight' has a level with the surface insertion of achenes on fruit and an inserted position of calyx attachment on fruit, while 'Driscoll Osceola' has a below the surface insertion of achenes on fruit and a level with fruit position of calyx attachment on fruit. Further, 'DrisStrawSeventyEight' is higher yielding than 'Driscoll Osceola'.

'DrisStrawSeventyEight' differs from the commercial variety 'DrisStrawThirtySeven' (U.S. Plant Pat No. 25,866) in that 'DrisStrawSeventyEight' has inflorescence above foliage, while 'DrisStrawThirtySeven' has inflorescence beneath foliage. Additionally, 'DrisStrawSeventyEight' has an obtuse shape of base of terminal leaflet and a convex shape in cross section of terminal leaflet, while 'DrisStrawThirtySeven' has a rounded shape of base of terminal leaflet and a slightly convex shape in cross section of terminal leaflet. Moreover, 'DrisStrawSeventyEight' has a cordate fruit shape and a partially remontant type of bearing, while 'DrisStrawThirtySeven' has a conical fruit shape and a not remontant type of bearing. Further, 'DrisStrawSeventyEight' is higher yielding than 'DrisStrawThirtySeven'.

We claim:

1. A new and distinct variety of strawberry plant named 'DrisStrawSeventyEight' as shown and described herein.

* * * * *

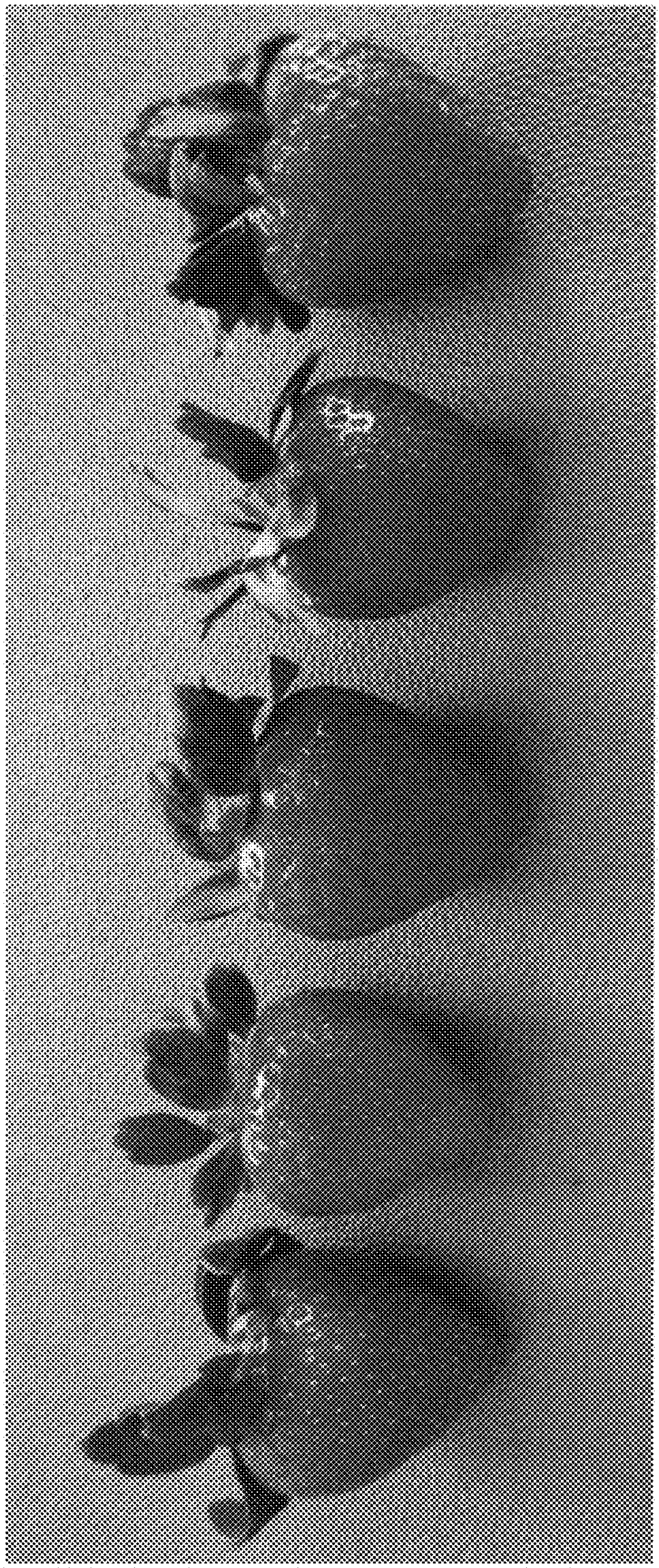

FIG. 1

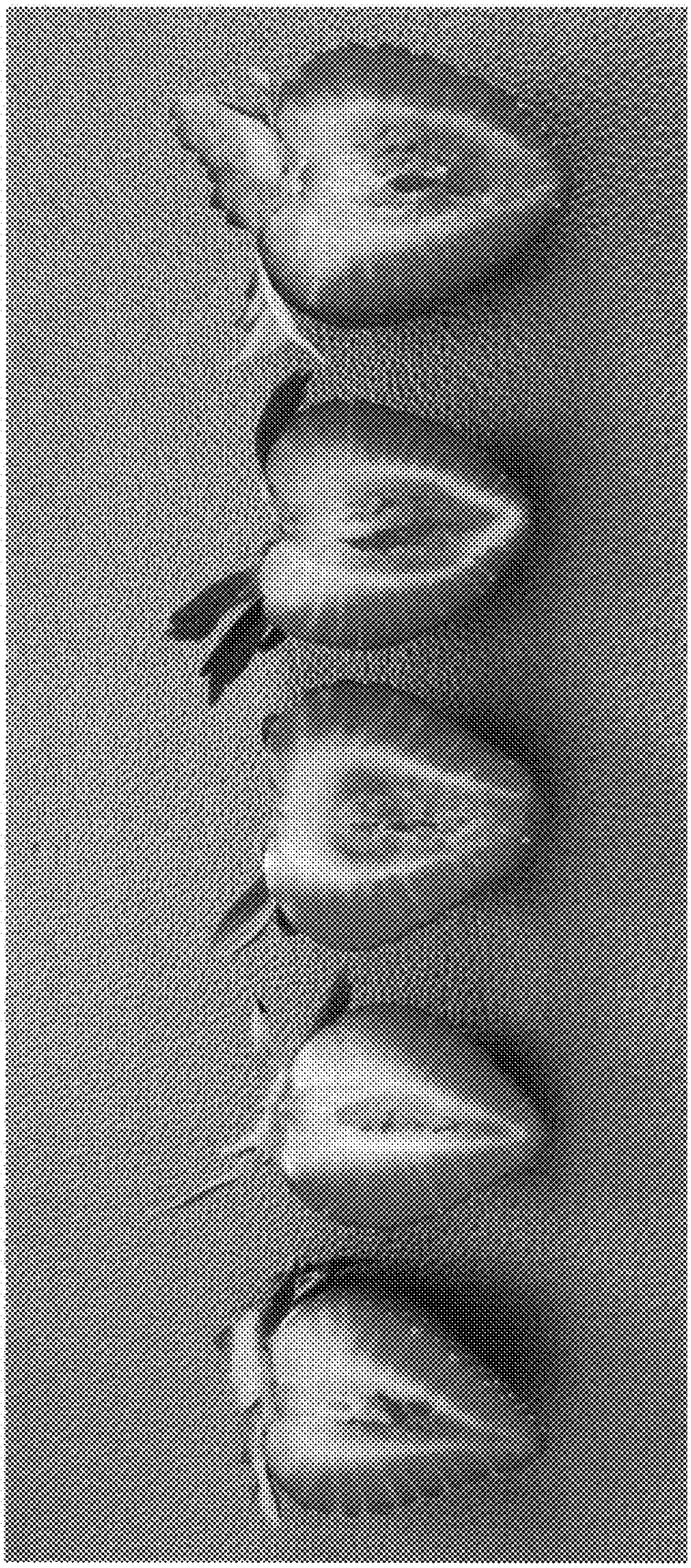

FIG. 2

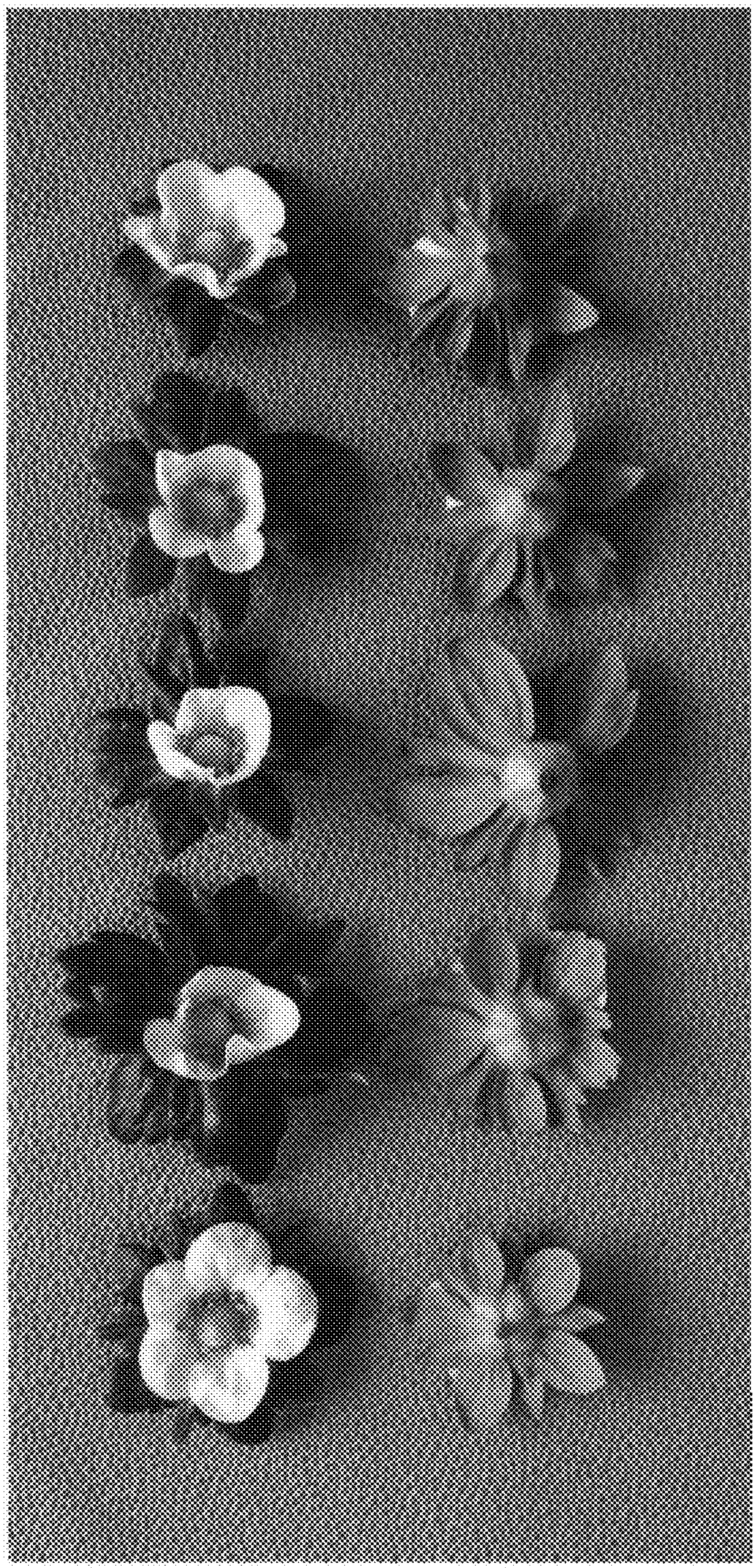

FIG. 3

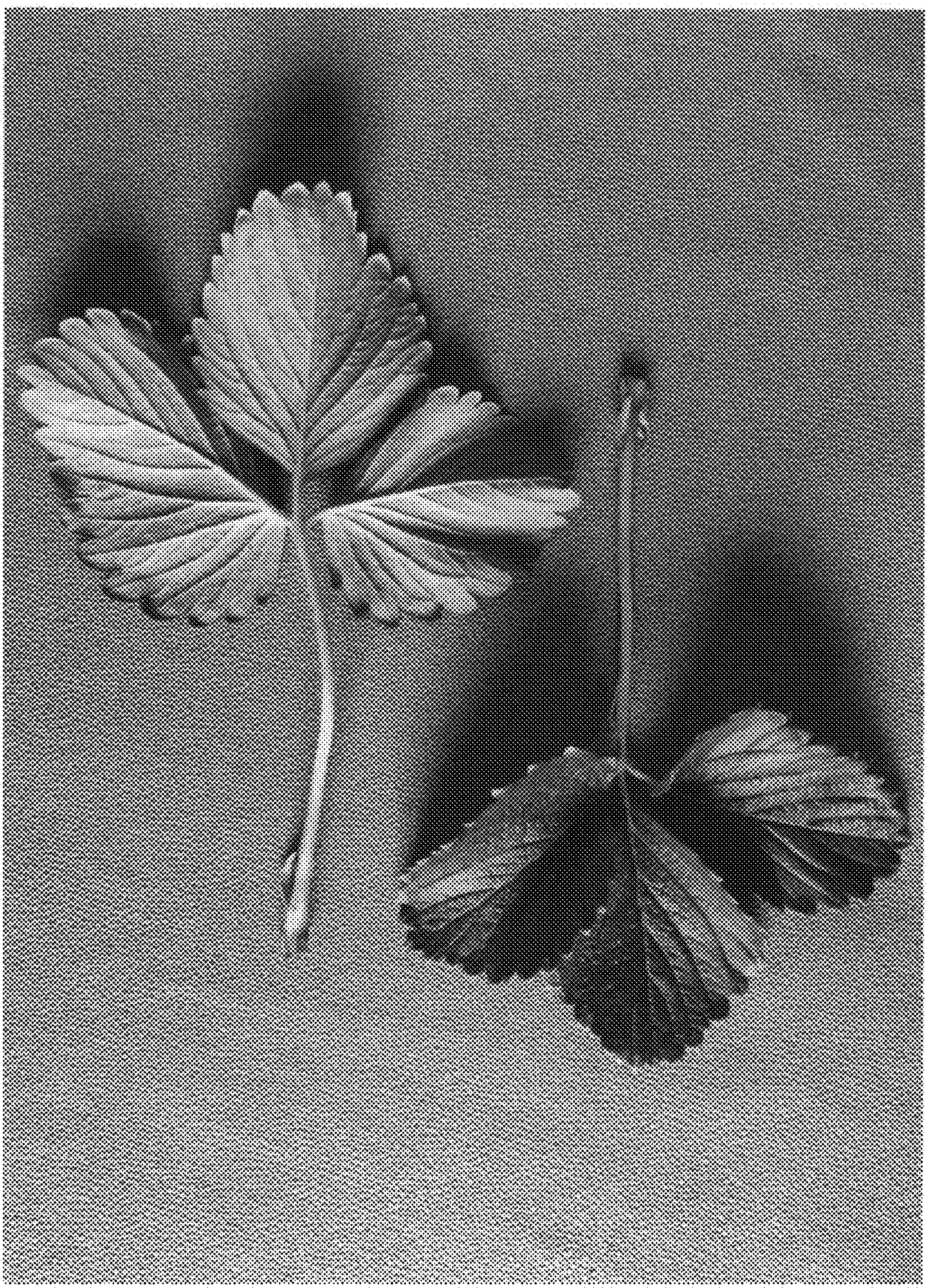

FIG. 4B

FIG. 4A

FIG. 5